

GCE AS

WJEC/EDUQAS

B520U20-1 (Practice Paper 2 - Advanced)

ECONOMICS – AS component 2

Exploring Economic Issues

2 hours

ADDITIONAL MATERIALS

In addition to this examination paper, you will need a calculator and a WJEC pink 16-page answer booklet.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen.

You may use a pencil for graphs and diagrams only.

Answer **all** questions.

Write your answers in the separate answer booklet provided.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

You are reminded of the necessity for good English and orderly presentation in your answers.

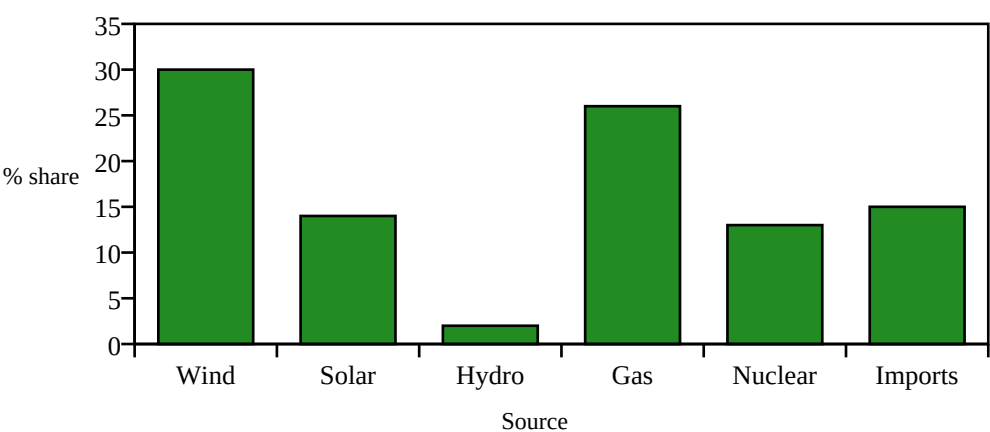
Answer **all** questions.

1. The Green Transition: Powering the UK to Net Zero

In 2019, the UK became the first major economy to legislate a legally binding target to reach net zero greenhouse gas emissions by 2050. Achieving this target requires a fundamental restructuring of the UK economy, with major investment in renewable electricity generation, electric vehicle (EV) infrastructure, low-carbon heating, energy-efficient buildings and emerging hydrogen technology. The Climate Change Committee estimates that capital investment of approximately £50bn per year by the early 2030s will be required.

As part of this strategy, the UK government has expanded the Contracts for Difference (CfD) scheme, which guarantees offshore wind operators a fixed price for electricity. It has also offered EV grants and zero VAT on solar panel installations. To raise revenue and internalise the negative externality of carbon emissions, the UK Emissions Trading Scheme (UK ETS) sets a price per tonne of CO2 emitted by heavy industry, currently around £40/tonne in 2024.

Figure 1: UK Electricity Generation Mix (% share by fuel source, 2024)



However, the green transition has been controversial. New offshore wind farms have drawn objections from coastal communities concerned about loss of marine habitats. Critics argue that the cost of net zero policies is regressive, with rising energy bills disproportionately affecting low-income households. The Office for Budget Responsibility estimates the total fiscal cost of net zero at over £344bn by 2050, though it also notes substantial economic gains from new green industries.

Table 1: Estimated impacts of UK net zero investment programme (2024–2030)

	Estimated Impact
Total green investment (public and private)	£300bn
Estimated net economic benefit (GVA over 25 years)	£420bn
Direct green jobs supported by 2030	480,000
Reduction in UK CO $\blacksquare$ emissions (vs 2019 baseline)	68% by 2030
Share of supply chain contracts to UK firms	60%
Households served by renewable electricity	21 million
Annual fall in household energy bills (long run, est.)	£300

Source: Adapted from Climate Change Committee, OBR and Department for Energy Security and Net Zero

Some economists argue that the UK ETS does not go far enough — only around 28% of UK emissions are covered. They propose introducing a comprehensive carbon tax that would cover all sectors, including household gas heating and road transport, with revenues recycled to households through 'carbon dividends' to address regressivity concerns. Meanwhile, others argue that command-and-control regulation (such as the 2030 ban on new petrol and diesel cars) would be a more reliable way to deliver emissions cuts.

- (a) Using a diagram, outline the likely impact of expanded offshore wind investment on the wages of skilled offshore engineers. **[4]**
- (b) Discuss the likely cross elasticity of demand between electric vehicles and petrol in the UK. **[6]**
- (c) Using **Table 1**, explain how the UK's green investment programme can lead to the benefits of the multiplier effect. **[6]**
- (d) Evaluate whether a comprehensive carbon tax is better than command-and-control regulation to reduce the negative externalities of carbon emissions in the UK. **[10]**
- (e) (i) Define the term 'government failure' and explain why renewable energy subsidies could be considered an example of government failure. **[4]**
(ii) Using the data, assess the extent to which the UK's net zero investment programme would be positive for the short run and long run macroeconomic performance of the UK. **[10]**

2. Funding the Green Transition

Following the 2008 Great Recession and the 2020 COVID-19 pandemic, the UK government took on significant additional debt. The Bank of England responded to the pandemic by reducing the Bank Rate to 0.1% (the lowest in its history) and launching a £450bn quantitative easing (QE) programme. The Treasury supported businesses and households through the £70bn furlough scheme and a temporary cut in VAT (an ad valorem tax) on hospitality from 20% to 5%.

Looking ahead, the UK faces the challenge of financing the green transition while managing high levels of public sector net debt. Several options have been proposed:

- A **carbon border adjustment mechanism (CBAM)**: a tax on imported goods from countries with weaker climate policies, raising revenue while preventing carbon leakage.
- A **green sovereign wealth fund**: £100bn fund invested in renewable energy infrastructure and emerging clean technologies, expected to generate long-run returns.
- A **frequent flyer levy**: progressive air passenger duty rising with each additional flight, targeting the heaviest emitters.
- Reform of **fossil fuel subsidies**: removing tax reliefs for North Sea oil and gas firms could raise £8bn per year.

Figure 1: UK Public Sector Net Debt as % of GDP (2008–2024)

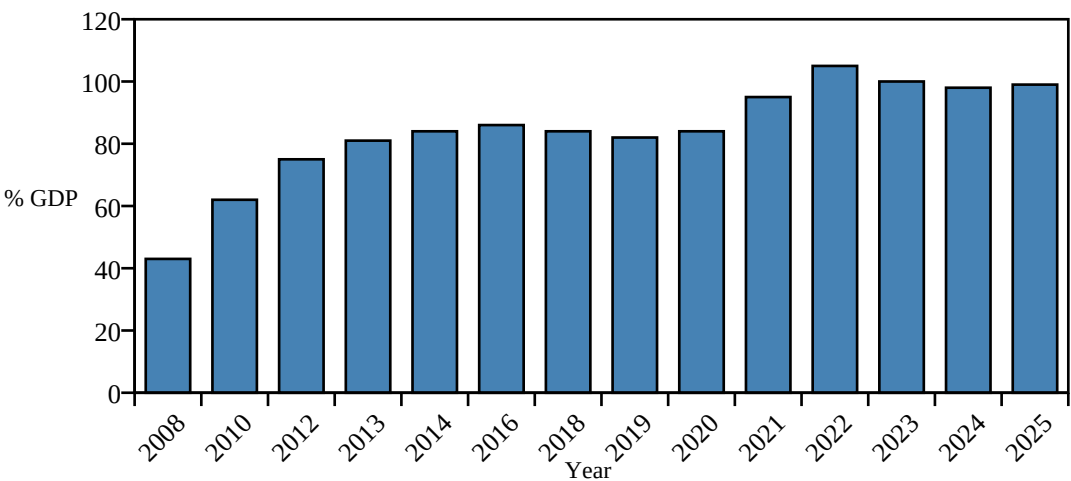


Table 1: Proposed UK green tax and revenue measures

Measure	Details
Carbon tax (extension)	Comprehensive tax of £75/tonne across all sectors by 2030; estimated to raise £18bn per year
Carbon border adjustment (CBAM)	Tax on carbon-intensive imports (steel, aluminium, cement); estimated to raise £2bn annually
Frequent flyer levy	Progressive air passenger duty: £0 for first flight, rising to £100+ per flight thereafter
Wealth tax on top 1%	1% annual tax on net wealth above £10m; estimated revenue: £10bn per year
Fossil fuel subsidy reform	Removal of tax reliefs for North Sea operators; raises £8bn per year

Critics warn that overly aggressive green taxation could harm UK competitiveness and trigger 'carbon leakage', where firms relocate to countries with weaker climate rules. Others argue that the regressive impact on low-income households (who spend a higher share of income on energy) requires careful design, such as tax-and-dividend or income-contingent rebates. There is also debate about whether targeted policies — such as free retraining for workers in carbon-intensive industries — would be more effective at reducing inequality than universal energy subsidies.

- (a) (i) Using a PPF diagram, illustrate the impact of an increase in green technological progress on an economy that produces consumer goods and capital goods. [2]
- (ii) Using an AD/AS diagram, explain how expansionary monetary policy (line 4) can be used to stimulate economic recovery from a recession. [6]
- (b) Using the data, describe how the proposed comprehensive carbon tax and green sovereign wealth fund could
- (i) decrease negative externalities [4]
- (ii) increase positive externalities [4]
- (c) Using a demand and supply diagram, consider who benefits the most from a decrease in ad valorem VAT on green technologies (such as solar panels), the consumer or the firms? [7]
- (d) Using the data from **Table 1**, discuss whether the proposed green tax and revenue measures are a good way to raise government revenue. [9]
- (e) Using the data, discuss whether worker retraining programmes or energy bill rebates is the best way to ensure a 'just transition' that reduces inequality during the move to net zero. [8]

END OF PAPER